loading at moderate pavement temperatures that might lead to fatigue cracking in an asphalt pavement. Because fatigue generally occurs at lower to moderate pavement temperatures after the pavement has been in service for a time, the specification addresses the response of the asphalt binder after aging in both the RTFO and PAV.

The DSR is again used to generate G^* and δ. However, instead of dividing the two parameters, the two are multiplied to produce a factor that may be related to the fatigue performance of an asphalt binder. The fatigue cracking factor is $G^*sin\ \delta$ (read as "G star sine delta"). It is the product of the complex shear modulus, G^*, and the sine of the phase angle, δ. The PG specification requires that all PG asphalt binders have a maximum value of 5000 kiloPascals for $G^*\sin\ \delta$ at the appropriate grade temperature. Low values of G^* and δ are considered desirable attributes from the standpoint of resistance to fatigue cracking. Thus, the PG specification promotes the use of compliant, elastic binders, after long-term (PAV) aging to address fatigue cracking.

Thermal Cracking

When temperature decreases, asphalt pavements shrink. As the temperature drops, the asphalt binder contracts to a much greater degree than the aggregate in an asphalt pavement, thus causing thermal stresses to develop in the pavement. When these stresses exceed the tensile strength of the asphalt mixture, a low-temperature crack develops. Because thermal cracking occurs at low pavement temperatures after the pavement has been in service for a time, the specification addresses the response of the asphalt binder after aging in both the RTFO and PAV.

In the PG specification, the main way of examining the propensity of an asphalt binder to develop thermal stresses at a specified low temperature is to use data generated from the BBR. If the stiffness is too high, the asphalt binder will behave in a brittle manner, indicating that cracking is more likely to occur. To minimize the contribution of the asphalt binder to thermal cracking, the creep stiffness *(S)* after 60 seconds of loading at the appropriate grade temperature must not exceed 300 megaPascals. In addition to stiffness, the rate at which the asphalt binder stiffness changes with time at low temperatures is regulated through the m-value. In the PG specification, a higher m-value is an indication that the asphalt binder may not increase in stiffness as rapidly when the temperature decreases and contraction occurs, leading to smaller tensile stresses in the asphalt binder and less chance for low temperature cracking. A minimum m-value of 0.300 after 60 seconds of loading at the appropriate grade temperature is required by the PG specification.

Past studies have also indicated that if a binder can stretch at least 1.0 percent of its original length during this thermal contraction period, cracks are less likely to occur. As a result, the direct tension test (DTT) is included in the PG specification as an alternative requirement to simply using creep stiffness and m-value. In the specification, the DTT requirement only applies to asphalt binders that have a creep stiffness greater

than 300 megaPascals and less than 600 megaPascals, with an m-value of 0.300 or greater. If the creep stiffness at the specified temperature is 300 megaPascals or less, then the direct tension test is not required. If the DTT is used, the failure strain must be a minimum of 1.0 percent at the appropriate grade temperature.

Although stiffness can also be used to estimate failure or strength properties, for some asphalt binders, especially modified asphalts, the relationship between stiffness and strength properties is not well known. This is why some researchers believe that the alternate procedure for determining a critical low cracking temperature specified in Table 2 of the AASHTO M 320 specification represents a more rigorous approach than the use of BBR stiffness and m-value alone (as is commonly used in the PG specification). In the alternate procedure, the BBR data is used at all loading times, and at two temperatures, to generate a predicted thermal stress curve as a function of temperature. The DTT is then used to determine the failure stress in an asphalt binder specimen when tensile fracture occurs. The failure tensile stress is plotted to determine at what temperature the thermal stress curve from the BBR data exceeds the failure stress from the DTT (see Figure 2.37). This intersection is called the "critical cracking temperature" and is used as an alternate method of determining the low temperature grade of an asphalt binder.

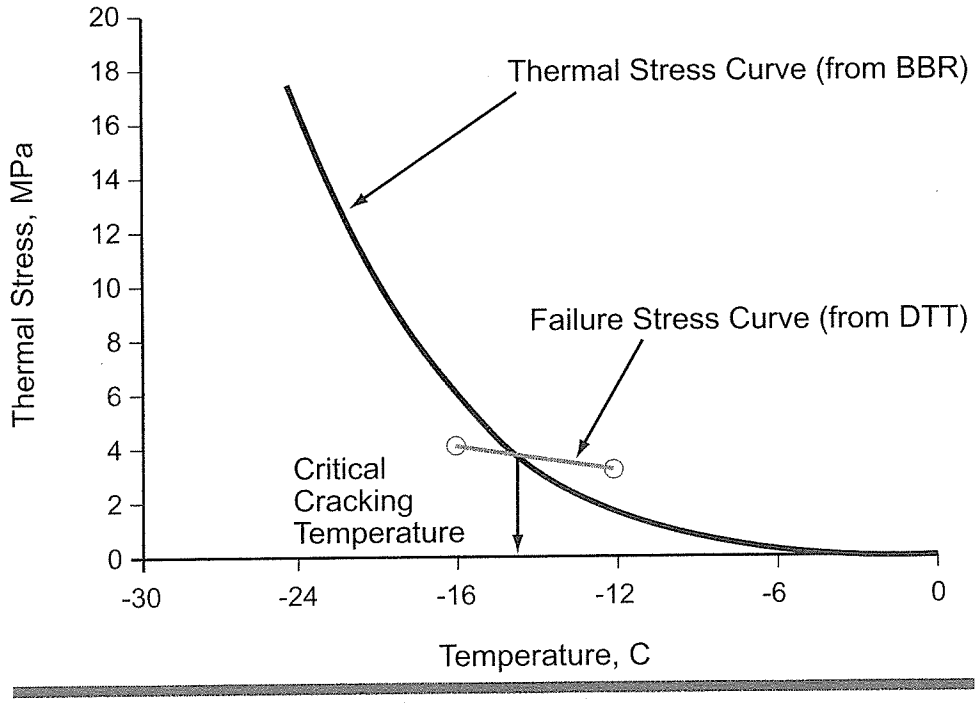


Figure 2.37 Thermal Stress Curves for Critical Cracking Temperature

2.4 Emulsified and Cutback Asphalts

2.4.1 Introduction

To use an asphalt cement for construction purposes, it must be liquefied before it can be pumped through pipes, mixed with aggregate, or sprayed through nozzles. Usually the liquification of asphalt cements occurs when the material is heated to elevated temperatures (e.g., 275°F or 135°C). When asphalt cement cools, it becomes a semi-solid cementing material.

There are other ways, however, to liquefy asphalt for construction operations. The asphalt binder can be dissolved in selected petroleum solvents or emulsified with an emulsifying agent and water. Asphalt binders produced by these methods are known as cutback and emulsified asphalts, respectively.

Once the cutback or emulsified asphalt is used in liquid form during the construction process, the solvent (cutback asphalt) or water (emulsified asphalt) will evaporate, leaving the asphalt cement residue to perform its function. Note that the evaporation of petroleum solvents into the atmosphere may be restricted or prohibited by environmental regulations. As a result, the use of cutback asphalts may not be as prevalent as emulsified asphalts.

The process for making emulsified and cutback asphalt materials is shown in the petroleum asphalt flowchart (see Figure 2.1). In each case, asphalt cement is the base material that is liquefied by either cutting the asphalt with a petroleum solvent or producing an emulsion of asphalt cement, water, and emulsifying agents.

Emulsified Asphalts

In the emulsification process, asphalt cement is mechanically milled into tiny globules and dispersed into water treated with a small quantity of emulsifying agent. In an emulsified asphalt, the water is the continuous phase and the globules of asphalt make up the discontinuous phase. The machine used in the emulsification process is a colloid mill, and the asphalt globules that are produced are extremely small, mostly in the colloidal size range (5–10 microns and smaller).

By proper selection of emulsifying agents and other manufacturing controls, emulsified asphalts are produced in several types and grades. By choice of emulsifying agent, the emulsified asphalt may be one of the following:

- Anionic—Asphalt globules are electronegatively charged.
- Cationic—Asphalt globules are electropositively charged.
- Non-ionic—Asphalt globules are neutral (have no charge).

In practice, the first two types have ordinarily been used in roadway construction and maintenance. Non-ionic emulsions, however, have become more widely used as emulsion technology has advanced.

Also, by variations in materials and manufacture, emulsified asphalts are made in several grades as shown in Table 2.3.

Table 2.3
Emulsified Asphalt Grades

	Anionic	Cationic
Rapid Setting (RS)	RS-1	CRS-1
	RS-2	CRS-2
	HFRS-2	
Medium Setting (MS)	MS-1	
	MS-2	CMS-2
	MS-2h	CMS-2h
	HFMS-1	
	HFMS-2	
	HFMS-2h	
	HFMS-2s	
Slow Setting (SS)	SS-1	CSS-1
	SS-1h	CSS-1h
Quick Setting (QS)	QS-1h	CQS-1h

In Table 2.3, the RS, MS, SS, or QS designation in the emulsified asphalt grade is an indication of the emulsion setting rate: rapid setting, medium setting, slow setting, or quick setting, respectively. The *h* that follows certain grades simply means that a harder base asphalt is used. The *HF* preceding some of the MS grades indicates that the material is a high-float emulsified asphalt, as measured by the Float Test (AASHTO T 50 or ASTM D 139). High-float emulsions have a quality, imparted by the addition of certain chemicals, which permits a thicker asphalt film on the aggregate particles with lessened probability of drainage.

Note that the list in Table 2.3 may not be all-inclusive. In addition, it is highly unlikely that a producer would carry an inventory of all the grades in Table 2.3. Users should check local supplies before specifying a specific grade of emulsified asphalt.

Emulsified asphalts that are used in construction operations must remain fluid enough for proper application. The asphalt globules are kept apart by the emulsifying agent until the emulsion has been placed on the surface of a soil, existing pavement, or aggregate particles. If the asphalt binder in the emulsion is to perform its ultimate function of cementing and waterproofing, the asphalt phase must separate from the water phase. This occurs through one or both of the following:

- Neutralization of the electro-static charges
- Water evaporation

When the phase separation of an emulsified asphalt begins, the emulsion is said to "break." Once the emulsion breaks, the asphalt droplets coalesce and produce a continuous film of asphalt on the aggregate or pavement. The speed of this coalescence of asphalt is related to the setting characteristics of the grade. As expected, coalescence occurs more quickly with the rapid-setting grades. The manner and rate at which emulsified asphalts break depends largely on the quantity and properties of the emulsifying agent used in manufacture of the emulsion, and on the relative proportion of water and asphalt.

Cutback Asphalts

Petroleum solvents used for dissolving asphalt cement are variously called distillate, diluent, or cutter stock. If the solvent used in making the cutback asphalt is highly volatile, it will quickly escape by evaporation. Solvents of lower volatility evaporate more slowly. Cutback asphalts are divided into the following three types based on the relative speed of evaporation:

- Rapid-curing (RC)—asphalt cement combined with a light diluent of high volatility, generally with a boiling point similar to gasoline or naphtha. Grades include RC-70, RC-250, RC-800, and RC-3000.
- Medium-curing (MC)—asphalt cement combined with a medium diluent of intermediate volatility, generally with a boiling point similar to kerosene. Grades include MC-30, MC-70, MC-250, MC-800, and MC-3000.
- Slow-curing (SC)—asphalt cement combined with oils of low volatility. Grades include SC-70, SC-250, SC-800, and SC-3000.

Slow-curing cutback asphalts may also be referred to as road oils. This term originated in earlier days when asphalt residual oil was used to give roads a low-cost, all-weather surface.

The degree of liquidity developed in each cutback asphalt depends principally on the proportion of solvent to asphalt cement. To a minor degree, the liquidity of the cutback may be affected by the hardness of the base asphalt from which the cutback is made. The more viscous grades of cutback asphalt are indicated by higher numbers in the grade designation and may require a small amount of heating to make them fluid enough for construction operations.

Uses of Emulsified and Cutback Asphalts

Emulsified asphalts have advantages over asphalt cements in that they can be used with cold as well as heated aggregate, and with aggregate that is dry or damp. Their ability to be used with damp aggregate gives emulsified asphalts an advantage over cutback asphalts as well.

Emulsified asphalts are used both for road construction and for many specialty applications. The rapid-setting grades are designed to react quickly with aggregate and revert from the emulsion state to asphalt. They are used primarily for spray applications, such as aggregate (chip) seals, sand seals, and similar surface treatments. The RS-2 and CRS-2 grades have a higher viscosity to minimize the potential for runoff.

The medium-setting grades are designed for mixing with coarse aggregate. Because these grades do not break immediately upon contact

with aggregate, mixes using them remain workable for a few minutes. They are used extensively in travel plants and mixed-in-place operations. The CMS grades have high viscosities to prevent runoff.

The slow-setting grades are designed for maximum mixing stability. They are used with dense-graded aggregate with a high fines content. The SS emulsions are generally used for dense-graded aggregate-emulsion bases, soil-asphalt stabilization, asphalt surface mixes, and slurry seals.

Quick-setting grades are specialty emulsions used for Quick Set Slurry Seal systems.

Cutback asphalts can be used with cold aggregates, with a minimum of heat. RC and MC types of cutback asphalt have been widely used in a variety of roadway, airfield, industrial, and specialty uses. Among the more important uses were road-mixing operations, stockpile mixes, and spray applications (such as prime, tack, and seal coats). But because the diluents in cutback asphalts are high-energy products that are lost by evaporation and because of environmental regulations, emulsified asphalts have largely supplanted cutbacks for most of these uses. SC cutback asphalts are used primarily in road-mixing and dust-laying applications. They are also used for stockpile patching mixes, plant-mixing with graded aggregates, and occasionally for priming. Here too, use is declining. It is suggested that local environmental regulations be reviewed before using a cutback asphalt.

2.4.2 Tests for Emulsified Asphalts

Saybolt Furol Viscosity

The Saybolt Furol viscosity test is used to measure the consistency (rate of flow) of emulsified asphalts. As a matter of testing convenience, and also to achieve suitable accuracy, the test is performed at one of two temperatures, 77°F or 122°F (25°C or 50°C) depending on the viscosity characteristics of the specific type and grade of asphalt emulsion.

For testing at 77°F (25°C), a sample is carefully stirred and conditioned for 30 minutes at the test temperature. The sample is then poured through a 0.85-millimeter sieve or strainer into a standard Saybolt Furol viscometer with a stopper placed in the orifice. The test is started by first withdrawing the stopper and determining the time required for 60 milliliters of asphalt to flow through the orifice (see Figure 2.38). The time interval is called Saybolt Furol viscosity and is measured in seconds. The more viscous the material, the greater the length of time required for a given volume to flow through the orifice. Thus, a higher Saybolt Furol time is indicative of a higher sample viscosity.

For testing at 122°F (50°C), the sample is first heated to 122±5°F (50±3°C) and then poured through the strainer into the viscometer and brought to test temperature before stopper is removed and the flow is timed, as already described.

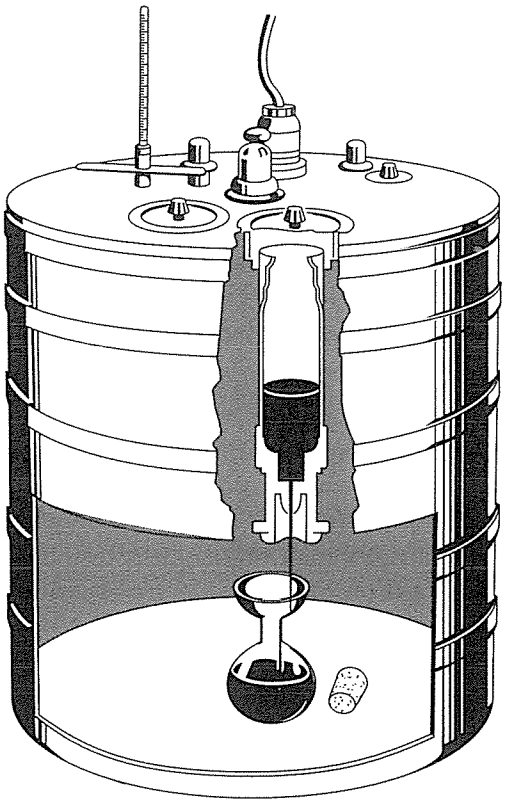


Figure 2.38 Saybolt Furol Viscosity

The Saybolt Furol viscosity test is suitable for testing emulsified asphalts that have a minimum flow time of 20 seconds.

Storage Stability and Settlement

The purpose of the storage stability and settlement tests is to detect the tendency of asphalt globules to "settle out" during storage of emulsified asphalts. The user is provided with an element of protection against separation of asphalt and water in unstable asphalt emulsions that may be stored for a period of time.

Either test is conducted by placing a 500-milliliter sample in a graduated cylinder, which is then stoppered and allowed to stand undisturbed. For the storage stability test, the sample must remain undisturbed at 25±3°C (77±5°F) for 24 hours. For the settlement test, the sample must remain undisturbed at 25±3°C for 5 days. At the end of the storage period, small samples are then taken from the top and the bottom parts of the cylinder. Each sample is placed in an individual beaker and weighed. The samples are then heated to 325°F (163°C) for a total of three hours, allowed to cool to room temperature, and weighed. The weights obtained provide the basis for determining the difference, if any, between asphalt cement content in the upper and lower portions of the graduated cylinder, thus providing a measure of the settling of the asphalt globules from the emulsion.

Classification

The classification test is a procedure for distinguishing rapid-setting cationic emulsified asphalt from other types by their failure to coat a sand-cement mixture.

The procedure requires that a washed and dried sample of Ottawa Sand (465 grams) be mixed in a beaker with a 35-gram sample of emulsified asphalt and thoroughly mixed for two minutes. At the end of the mixing period, any excess emulsion is drained off and the mix is placed on absorbent paper for visual examination. An excess of uncoated area (compared with the coated area of sand particles) is considered a passing rating for rapid-setting cationic emulsions.

Demulsibility

The demulsibility test is used to identify rapid-setting emulsified asphalts by indicating the relative rate at which the colloidal asphalt globules will break when spread in thin films on soil or aggregate. Calcium chloride causes the minute asphalt globules present in these emulsified asphalts to coalesce. In the test, a solution of calcium chloride and water is thoroughly mixed with emulsified asphalt and then decanted over a sieve to determine how much the asphalt globules coalesce.

In rapid-setting (RS) emulsions tests, a very weak solution of calcium chloride and water is used. Specifications prescribe the concentration of the solution and the minimum amount of asphalt to be retained on the sieve. A high degree of "demulsibility" indicates an RS emulsion, which is expected to break almost immediately upon contact with the aggregate on which it is applied.

When cationic rapid-setting (CRS) emulsions are tested, a solution of sodium dioctyl sulfosuccinate in water is used in place of a calcium chloride solution.

Cement Mixing

The cement-mixing test is used instead of the demulsibility test for slow-setting (SS) grades of emulsified asphalt. This test is specified for both the anionic and cationic types to ensure that the products are substantially immune from a rapid coalescence of asphalt particles in contact with fine-graded soils or dusty aggregates.

The cement-mixing test is performed by stirring together 100 milliliters of emulsion—which has been diluted with water to 55 percent residue—with 50 grams of high-early-strength Portland cement (Type III). After these ingredients are stirred for one minute, 150 milliliters of water is added and the mixture is stirred for an additional three minutes. The mixture is then washed over a 1.40 millimeter (No. 14) sieve and the percent of material retained on the sieve is determined. A low amount of retained weight (i.e., particles that have "broken") is desirable for SS grades.

Coating Ability and Water Resistance

This test has a threefold purpose. It determines the ability of an asphalt emulsion to coat aggregate thoroughly, to withstand mixing action while

remaining as a film on the aggregates, and to resist the washing action of water after completion of mixing. The test is primarily intended to identify medium-setting asphalt emulsions suitable for mixing with coarse-graded calcareous aggregates. Other aggregates may be used in the test if calcium carbonate is omitted throughout the method. This test is not adaptable to rapid-setting or slow-setting asphalt emulsions.

The test is conducted by coating 461 grams of a standard reference aggregate with 4 grams of calcium carbonate dust. The combined aggregate is then mixed with 35 grams of emulsified asphalt for five minutes. About half of the mixture is then placed on absorbent paper for a visual inspection of the surface area of aggregate coated by the emulsified asphalt. The remainder of the mixture is sprayed with water and rinsed until the rinse water runs clear. This material is then placed on absorbent paper and inspected for coating retention.

The procedure is then repeated using wet aggregate. Visual inspections are made as previously described for the dry aggregate tests.

Particle Charge

The particle charge test is used to identify cationic asphalt emulsions. The test is conducted by heating a sample of emulsion to 122°F (50°C) and pouring it into a 250-milliliter beaker. A positive electrode (anode) and a negative electrode (cathode) are then immersed in the sample and connected to a controlled direct-current electrical source providing a current of at least 8 milliamperes. After 30 minutes, or after the current has dropped to 2 milliamperes, the two electrodes are examined. An asphalt deposit on the cathode that is clearly discernible (compared with the anode) indicates that the sample is a cationic-emulsified asphalt.

Note that this test may not produce conclusive results for CSS grades. If the test is inconclusive, then a mixing test using silica sand is recommended (as described in ASTM D 244).

Sieve

The sieve test complements the settlement test and has a somewhat similar purpose. It is used to determine quantitatively the percent of asphalt cement present in the form of pieces, strings, or relatively large globules. Such particles of asphalt might clog equipment and tend to provide nonuniform coatings of asphalt on aggregate particles. This nonuniformity might not be detected by the settlement test, which is of value in this regard only when there is a sufficient difference in the specific gravity of asphalt and water to allow settlement.

The test is conducted by bringing 1000 grams of asphalt emulsion to the proper test temperature depending on its Saybolt Furol viscosity. Emulsified asphalts having a high viscosity (greater than 100 seconds at 77°F (25°C) or any emulsion required to be tested at 50°C) are tested at 50°C. The remainder are tested at ambient temperature. The sample is then poured through an 850 microns (No. 20) sieve. For anionic emulsified asphalts, the sieve and retained asphalt are then rinsed with a mild sodium oleate solution. For cationic emulsified asphalts, distilled water is the rinse agent. After rinsing, the sieve and

asphalt are dried in an oven and the relative amount of asphalt retained on the sieve is determined. A low percentage of retained material is desirable in this test.

Distillation

The distillation test is used to determine the relative proportion of asphalt cement and water in the asphalt emulsion. Some grades of emulsified asphalt also contain an oil distillate; the distillation test provides information on the amount of this material in the emulsion. Also, the distillation test provides an asphalt cement residue, on which additional tests (penetration, solubility, and ductility) may be made, as previously described for asphalt cement.

As with the sieve test, emulsified asphalts with a high viscosity are heated to 50°C before beginning the procedure. In the distillation procedure, a 200 gram sample of emulsion is placed in an aluminum-alloy still and heated (see Figure 2.39). Once the temperature reaches 500°F (260°C), distillation is continued for 15 minutes to produce a smooth, homogeneous residue, which may then be used for further tests.

Rapid- and medium-setting grades of cationic asphalt emulsion may include an oily distillate, the maximum amount of which usually is limited by specifications. The distillate collected in the graduated cylinder includes both oil and water from the asphalt emulsion. Because these two materials separate in the graduated cylinder, the amounts of each can be determined.

Tests on Residue

After distillation, the asphalt residue may be tested using standard asphalt binder tests. As of this printing, standard specifications for emulsified

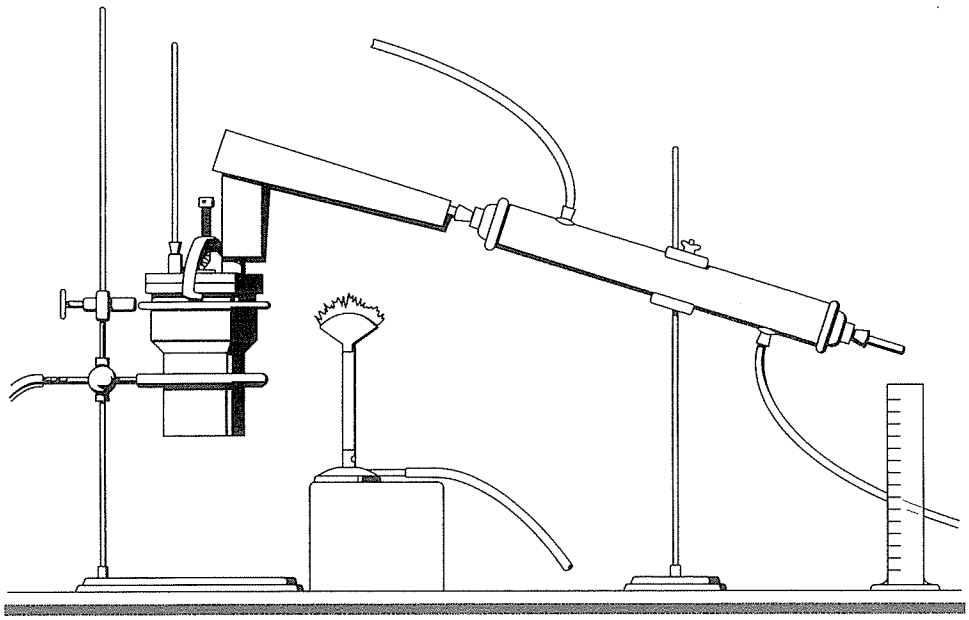


Figure 2.39 Distillation (Emulsion)

asphalt residue still use tests from the viscosity and penetration grading systems rather than from the PG asphalt binder system. These tests may include penetration, ductility, and solubility.

Float Test

One additional test is commonly conducted on residue produced from high float (HF) emulsions. The float test uses a semispherical float made from aluminum or aluminum-alloy that has an 11 millimeter diameter opening in the bottom of the float. The residue sample is heated and poured into a brass collar designed to be screwed in-place in the float. After the sample cools in the collar, it is placed in the float and allowed to float in a water bath operating at 140°F (60°C). The time is measured from the instant the float is placed in the bath until water breaks through the asphalt residue in the float, causing it to sink.

2.4.3 Specifications for Emulsified Asphalts

Emulsified asphalts, which are produced using a variety of asphalt cement bases and setting properties, are manufactured for a number of uses. Specifications for anionic and cationic asphalt emulsions are included in AASHTO and ASTM standards. For anionic emulsified asphalts (ASTM D 977), there are four standard types available: rapid-setting (RS), medium-setting (MS), slow-setting (SS), and quick-setting (QS). The same setting types also are available for cationic emulsified asphalts (ASTM D 2397) and are designated CRS, CMS, CSS, and CQS.

Five grades of high-float anionic emulsions—designated as HFRS-2, HFMS-1, HFMS-2, HFMS-2h, and HFMS-2s—are also part of the ASTM D 977 specification. There are no corresponding high-float cationic emulsions in standard specifications. High-float emulsions have a specific quality that permits a thicker film coating with a reduced risk of runoff.

Several tests are required to determine specification compliance of emulsified asphalts. All grades of emulsified asphalt have requirements on Saybolt Furol viscosity, storage stability (except for QS grades), sieve, and percentage of residue. The emulsified asphalt residue also has requirements for all grades on penetration, ductility, and solubility.

In addition to the above tests common for all emulsified asphalt grades, all cationic emulsified asphalt grades require a positive result on the particle charge test.

Additional tests, depending on the emulsified asphalt grade, are included in the specifications. For example, the demulsibility test is required only for rapid-setting (RS) grades. The coating ability and water resistance tests are only required for medium-setting (MS) grades. The cement-mixing test is required only for slow-setting (SS) grades.

These tests are listed in Table 2.4. Some of the tests for asphalt cement are also required for testing emulsified asphalt residue.

Table 2.4
Specification Tests for Emulsified Asphalt Grades

	AASHTO M 140 / ASTM D 977						AASHTO M 208 / ASTM D 2397			
Test	RS	HFRS	MS	HFMS	SS	QS	CRS	CMS	CSS	CQS
Viscosity, Saybolt	x	x	x	x	x	x	x	x	x	x
Storage Stability	x	x	x	x	x		x	x	x	
Sieve	x	x	x	x	x	x	x	x	x	x
Distilled Residue %	x	x	x	x	x	x	x	x	x	x
Demulsibility	x	x					x			
Coating Ability and Water Resistance			x	x				x		
Cement Mixing					x				x	
Oil Distillate %				x[1]			x	x		
Particle Charge							x	x	x	x
Residue Tests										
Penetration	x	x	x	x	x	x	x	x	x	x
Ductility	x	x	x	x	x	x	x	x	x	x
Solubility	x	x	x	x	x	x	x	x	x	x
Float Test		x		x						

[1] HFMS-2s only

2.4.4 Tests for Cutback Asphalts

Kinematic Viscosity

The kinematic viscosity test is used as a basis for classifying cutback asphalts into standard grades. The test procedure is similar to the kinematic viscosity test used for asphalt binders, except that the test is conducted in a thermostatically controlled water bath operating at 140°F (60°C).

Tag Open Cup Flash Point

The purpose and significance of the Tag Open Cup Flash Point test is the same as the Cleveland Open Cup Flash Point test described previously. The Tag Open Cup test is used for materials having flash points less than 199°F (93°C). This includes RC and MC grades, but not SC grades, which use the Cleveland Open Cup test.

The Tag Open Cup Flash Point procedure is essentially the same as the Cleveland Open Cup procedure, except that indirect heating is used because of the volatile nature of the diluent used in these products. The Tag Open Cup apparatus (see Figure 2.40) is glass, rather than metal, and is heated in a water bath rather than by direct flame.

Cutback asphalts commonly are used at temperatures above their flash point. Some rapid-curing cutbacks may flash as low as 80°F (27°C). The more volatile the solvent the cutback asphalt contains, the more hazardous its use. All these materials present some danger in use and must be handled properly.

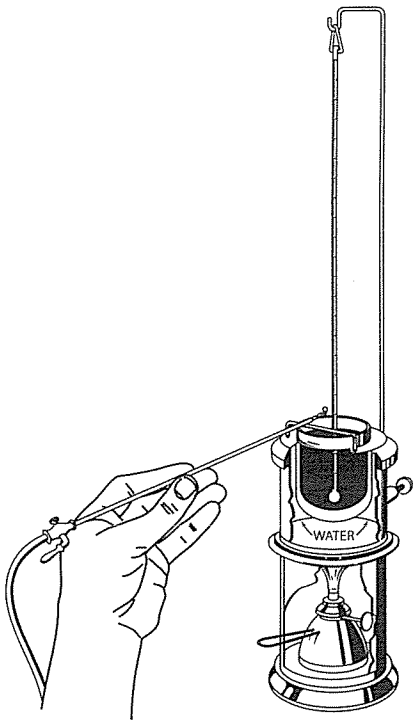


Figure 2.40 Flash Point (Tag Open Cup)

Water

Asphalt materials, except for emulsified asphalts, are usually specified to be substantially water free, primarily because water causes foaming when the material is heated, creating a potentially hazardous condition.

To determine the amount of water, if any, in cutback asphalt, a measured volume of asphalt is poured into a glass or metal still and thoroughly mixed with a xylene or high-boiling-range petroleum naphtha. A reflux condenser is attached to the still and discharged into a graduated trap. Heat is applied to the still and any water present is collected in the trap. The percentage of water by volume is then calculated.

Distillation

As previously indicated, cutback asphalts are blends of asphalt cement and suitable diluents. Properties of these materials are of importance in their application and performance.

The asphalt cement and diluent are separated by the distillation test (see Figure 2.41) to determine their quantity and for further identification. Approximately 200 milliliters of cutback asphalt is poured into a distillation flask connected to a water-cooled condenser tube. As the flask is slowly heated, the diluent vaporizes in the distillation flask and is again liquefied in the condensing tube. The condensate is drained into a graduated cylinder. The quantity of condensate driven off at several specified temperatures is

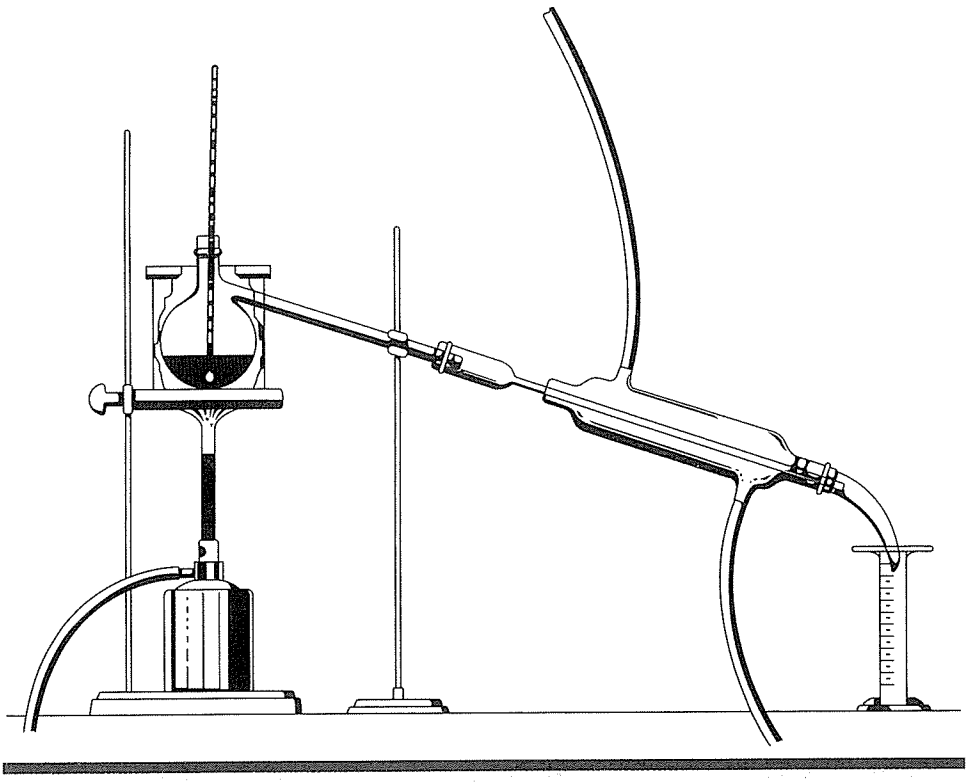


Figure 2.41 Distillation (Cutback)

determined, and this is an indication of the volatility characteristics of the diluent. After reaching 680°F (360°C), the material remaining in the distillation flask is considered asphalt cement residue.

The distillates that evaporate at various temperatures are of little importance for SC cutback asphalts. For these grades, the distillates that evaporate below the specified end point, 680°F (360°C), are largely of an oily nature, meaning that their rate of evaporation in service is quite slow. Therefore, only the total quantity of distillate driven off up to 680°F (360°C) is measured.

Tests on Residue

As with emulsified asphalt grades, the asphalt residue remaining after distillation may be tested using standard asphalt binder tests. As of this printing, standard specifications for cutback asphalt residue still use tests from the viscosity and penetration grading systems rather then the PG asphalt binder system. These tests may include viscosity, penetration, ductility, and solubility.

Residue of 100 Penetration

This test, described in ASTM D 243, is conducted only on SC cutback grades, and is perhaps the least significant to the consumer of all tests normally used on these materials. Because the in-service rate of curing of an SC cutback asphalt is quite slow, it may or may not reach a penetration of

100 during its service life. The principal value of the test is that it provides a residue upon which standard tests for asphalt binders may be made.

The test is performed by rapidly heating a small sample (100 grams) of SC cutback asphalt from 480°F to 500°F (249°C to 260°C) and maintaining it at this temperature until the operator judges that the material has achieved a penetration of 100 deci-millimeter. The weight of asphalt residue remaining is determined and the residue percentage is calculated.

2.4.5 Specifications for Cutback Asphalts

Cutback asphalts, which are produced using a variety of asphalt cement bases and diluents, are manufactured for a number of uses. AASHTO and ASTM have specifications for three standard types of cutbacks: rapid-curing (RC), medium-curing (MC), and slow-curing (SC). ASTM D 2026, D 2027, and D 2028 provide standard specifications for SC, MC, and RC cutbacks, respectively. In the specifications, the type (RC, MC, or SC) indicates the relative speed of evaporation. The grades (70, 250, 800, and 3000) indicate the minimum allowable kinematic viscosity in centistokes at 140°F (60°C). One additional grade, MC-30, has been used as a special grade for prime-coat applications in some sections of the United States. The maximum allowable viscosity for each grade is twice the minimum allowable value.

The most viscous grades (RC-3000, MC-3000, and SC-3000) are only moderately less viscous than the lowest viscosity-graded asphalt binders. The least viscous grades (RC-70, MC-30, MC-70, and SC-70) may be readily poured at room temperature.

Several tests are required to determine specification compliance of cutback asphalts. These tests are shown in Table 2.5 and are substantially the same for all cutback types and grades. The only exceptions are SC cutback asphalts that use the Cleveland Open Cup (COC) Flash Point test instead of the Tag Open Cup (TOC) Flash Point test, which is used by RC and MC cutback asphalts. SC cutbacks also have an additional test (the 100 Pen Residue test) that is not common to the other grades.

Table 2.5
Specification Tests for Cutback Asphalt Grades

	Method		Specifications[1]		
Test	AASHTO	ASTM	RC	MC	SC
Viscosity @ 60°C	T 201	D 2170	x	x	x
Flash Point, TOC	T 79	D 3143	x	x	
Flash Point, COC	T 48	D 92			x
Distillation	T 78	D 402	x	x	x
Residue Tests					
Viscosity @ 60°C	T 202[2]	D 2171[2]	x	x	x
Ductility	T 5	D 113	x	x	x
Solubility	T 44	D 2042	x	x	x
100 Pen Residue		D 243			x
Water %	T 55	D 95	x	x	x

[1] Cutback asphalt specifications are ASTM D 2026, D 2027, and D 2028.

[2] For SC cutback asphalts, substitute T 201 or ASTM D 2170.

2.5 Air-Blown Asphalts

2.5.1 Introduction

Special properties may be imparted to asphalt binders by blowing air through the topped crude fraction during the latter part of the refining process. The regular distillation process is discontinued at some point while the topped crude is still liquid. The heavy fraction is then put into a converter and air is blown through it while it is maintained at a high temperature. This process is continued until the asphalt has attained the desired properties. Often, such asphalts are called oxidized asphalts. However, this is not strictly a true term because not only oxidation but also vaporization, dehydrogenation, condensation, polymerization, and other reactions occur during the air-blowing process.

One of the most significant properties of air-blown asphalt is that the temperature at which asphalt becomes soft is raised. This temperature is called the softening point. Even at this temperature, asphalt still retains its waterproofing properties and durability. However, there is some reduction in ductility (when compared with asphalt cement for paving purposes).

Blown asphalts are used for a wide variety of industrial and specialty purposes, including roofing applications, pipe-coating enamels, undersealing applications (to fill cavities beneath Portland cement concrete pavements), and waterproofing membranes for lining canals and reservoirs. In many instances, blown asphalts are applied in pure form. In other cases, very fine mineral powders (fillers) are mixed with blown asphalt before application.

2.5.2 Properties, Specifications, and Tests

Properties and Specifications

Although similar in many respects to the asphalt binder grades used for paving, the blowing process provides materials that soften at higher temperatures than asphalt cements. Because the higher softening point is a most important and desirable property of blown asphalts, they are usually classified in terms of the ring-and-ball softening point test, rather than viscosity or penetration.

While blown asphalts are graded on the basis of the softening point, there are still penetration test requirements at three temperatures: 32°F, 77°F, and 115°F (0°C, 25°C, and 46°C). These requirements provide a degree of control over the temperature susceptibility, or the rate of consistency change with temperature, for these materials. Applications for air-blown asphalt and associated specifications include the following:

- Roofing—ASTM D 312
- Dampproofing and Waterproofing—ASTM D 449
- Canal, Ditch, and Pond Lining—ASTM D 2521
- Undersealing—ASTM D 3141

Tests

Ring-and-Ball Softening Point

The Ring-and-Ball softening point test is used as the basic measurement of consistency for grading blown asphalts. Figure 2.42 shows the test apparatus.

In the test, described in ASTM Test Method D 36, asphalt samples are confined in small brass rings and loaded with a steel ball in the center of each ring. After the samples are prepared, the assembly is suspended in a beaker of water, glycerine, or ethylene glycol at 1 inch (25 millimeters) above a metal plate. The liquid is then heated at a prescribed rate. As the asphalt softens, the balls and the asphalt gradually sink toward the plate. At the moment the asphalt touches the plate, the temperature of the water is determined, and this is designated as the Ring-and-Ball (R&B) softening point of the asphalt binder sample.

Penetration

For air-blown asphalts, the penetration test is conducted at three test temperatures as part of the specification. The test is the same procedure as that used for paving asphalt binders except that the total load and testing

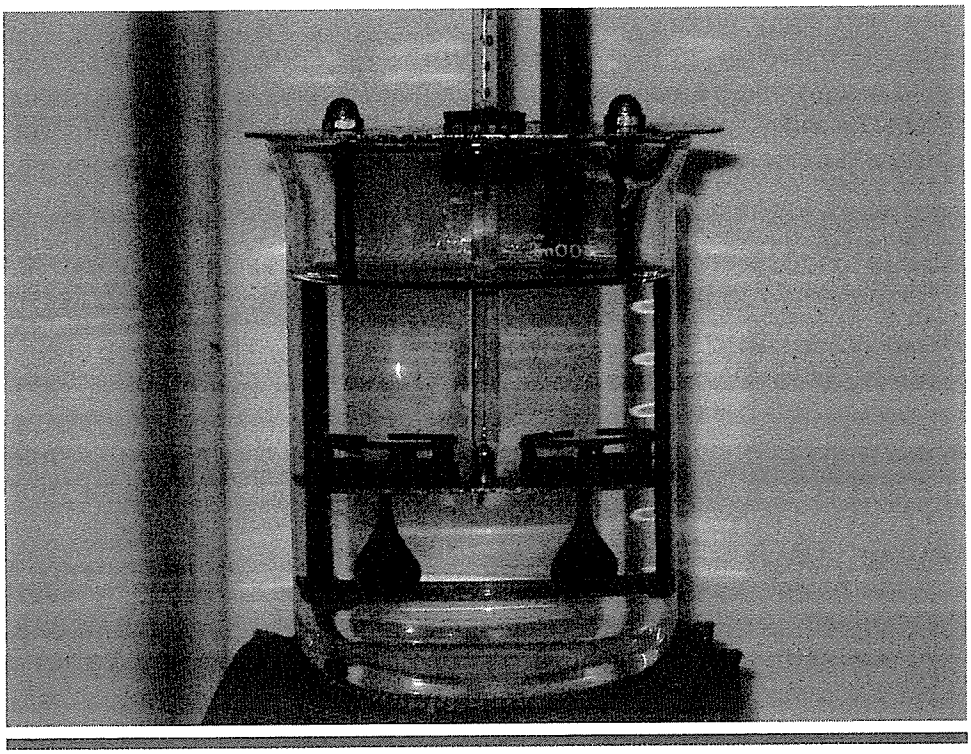


Figure 2.42 Ring-and-Ball Softening Point

time varies. At the standard test temperature of 77°F (25°C), a total load of 100 grams is used for five seconds. At the higher test temperature of 115°F (46 °C), a 50-gram load is used for five seconds. At the low test temperature of 32°F (0°C), a total load of 200 grams is used for 60 seconds.

Loss-on-Heating

The loss-on-heating test is generally similar to the thin film oven test as previously described. The only differences are in the dimensions of the asphalt sample. Whereas the asphalt sample in the TFO test is about 5.5 inches (140 millimeters) in diameter and ⅛ inch (3 millimeters) deep, the sample for the loss-on-heating test is approximately 2.2 inches (55 millimeters) in diameter and 1.4 inches (35 millimeters) deep. In both tests, the asphalt and container are placed on a rotating shelf in a ventilated oven and maintained at 325°F (163°C) for five hours. The shelf rotates at approximately five to six revolutions per minute.

As with the TFO test, the loss-on-heating also conditions asphalt for other tests. It subjects the asphalt to hardening conditions simulative of the application processes. A penetration test usually is made on the air-blown asphalt binder sample after the loss-on-heating test for comparison with the penetration of the asphalt before the test.

2.6 Asphalt Modification and Additives

2.6.1 Reasons for Using Additives and Modification

The modification of neat asphalt binder to enhance its performance characteristics has occurred in the United States for more than 50 years. In a recent survey of 20 experts representing 18 states, 70 percent responded that there was a definite benefit in using polymer-modified asphalt mixtures to extend the pavement's service life. Nearly 60 percent of those experts also responded that the use of polymer-modified asphalt mixtures significantly reduced maintenance costs, but there was insufficient performance data to quantify that benefit or enhancement.

The survey also indicated that the primary reason why users chose to use polymer-modified asphalt was to increase the mixture's resistance to rutting. Secondary reasons were to increase resistance to thermal cracking and to increase durability of the mixture.

It is important to note that while some user agencies have indicated a willingness to increase service life or reduce the risk of the development of early distresses through the use of modified asphalt binders (and their higher initial construction costs), not all asphalt pavements need to be constructed using modified asphalt mixtures. Each project should be evaluated to determine if the environmental conditions, traffic loading, expected service life and performance warrant the use of modified asphalt materials.

2.6.2 Types of Modifiers and Additives

Modified asphalt binders are sometimes generically referred to as "polymer-modified asphalts" or PMAs. In actuality, though, modified asphalt binders may be produced in a number of ways, including polymer and chemical modification, although polymer modification is the most prevalent.

Polymers are made up of smaller molecules, or monomers, that are chemically connected to form long molecular chains. Names of polymers are usually based on the names of the monomers used to make the specific polymer. Thus, the polymer that comes from the polymerization of ethylene monomers is called "polyethylene."

Most often, when monomers are polymerized, they form long-chain polymers. Some types of these polymers, called thermoplastics, soften when heated, enabling them to be re-formed. Some examples of thermoplastic polymers include nylon, polyethylene, and polyethylene terephthalate (which is used in clear plastic bottles).

By contrast, some monomers, when polymerized, connect to form a three-dimensional network. These types of polymers, called thermosets, are permanently set in shape. Some examples of thermoset polymers include epoxy resins and vulcanized rubber.

Two broad classes of polymers used in asphalt modification are polyolefins and styrenic polymers. Polyolefins are so named because they are based on the polymerization of molecules containing a simple double bond or olefin. Polyolefin polymers include polyethylene, polypropylene, and ethylene vinyl acetate. Styrenic polymers are so named because they are based on polystyrene that has been co-polymerized with other small molecules—most commonly butadiene.

Polymers can also be classified based on physical properties. Depending on their behavior when stretched with sufficient force, polymers are classified as plastomers (plastics) or elastomers (rubbers or elastics). When stretched, plastomers will yield and remain in their stretched position when the load is released. Elastomers will yield under load (stretch) but will return to their original shape when the load is released. Generally speaking, most polyolefins behave as plastomers, while styrene-butadiene copolymers behave as elastomers.

Finally, when blended into asphalt, polymers can behave in two different ways. If the polymer forms discrete particles in the asphalt binder, then it functions primarily as a thickener, which increases the viscosity of the asphalt binder while having no significant effect on low temperature properties. If the polymer forms a continuous network in the asphalt binder, then it functions as a homogeneous blend, which may impart some of the physical characteristics of the polymer to the asphalt binder. In this instance the high- and low-temperature properties of the asphalt binder may be affected.

Elastic Modification (Elastomers)

The most common co-monomer used to produce styrenic polymers is butadiene. When styrene and butadiene are polymerized in a random arrangement, referred to as a "random copolymer," the polymer is called styrene-butadiene rubber (SBR). This type of polymer may also be referred to as a synthetic latex. Figure 2.43 provides a microscopic view of SBR modification of asphalt.

When styrene and butadiene are polymerized in discrete, connected blocks, referred to as a "block copolymer," the polymer is called styrene-butadiene (SB) or styrene-butadiene-styrene (SBS). Figure 2.44 shows SBS in the form of pellets.

Although some elastomers are blended in-line at the hot mix asphalt mixing facility, the most common practice is to pre-blend the polymer and asphalt binder at the terminal or refinery (see Figure 2.45). Some PMA systems require only mixing of the polymer into the asphalt binder in a mixing kettle, while others require high shear milling or other special mixing operations. These processes take place at specific temperatures and are continued until the materials are appropriately mixed.

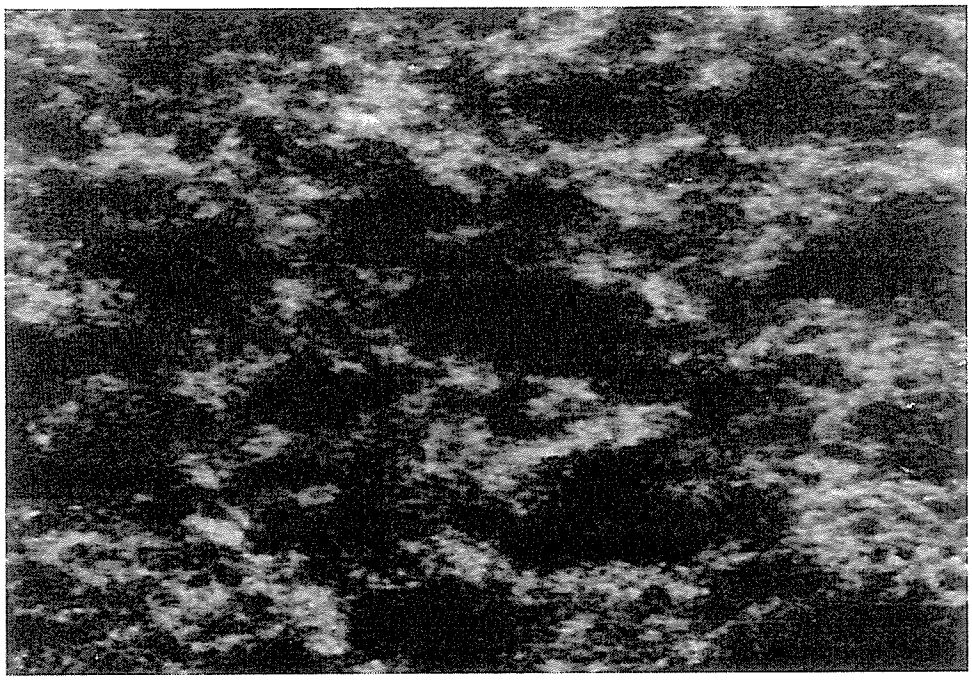


Figure 2.43 SBR Latex Modification of Hot Asphalt Cement

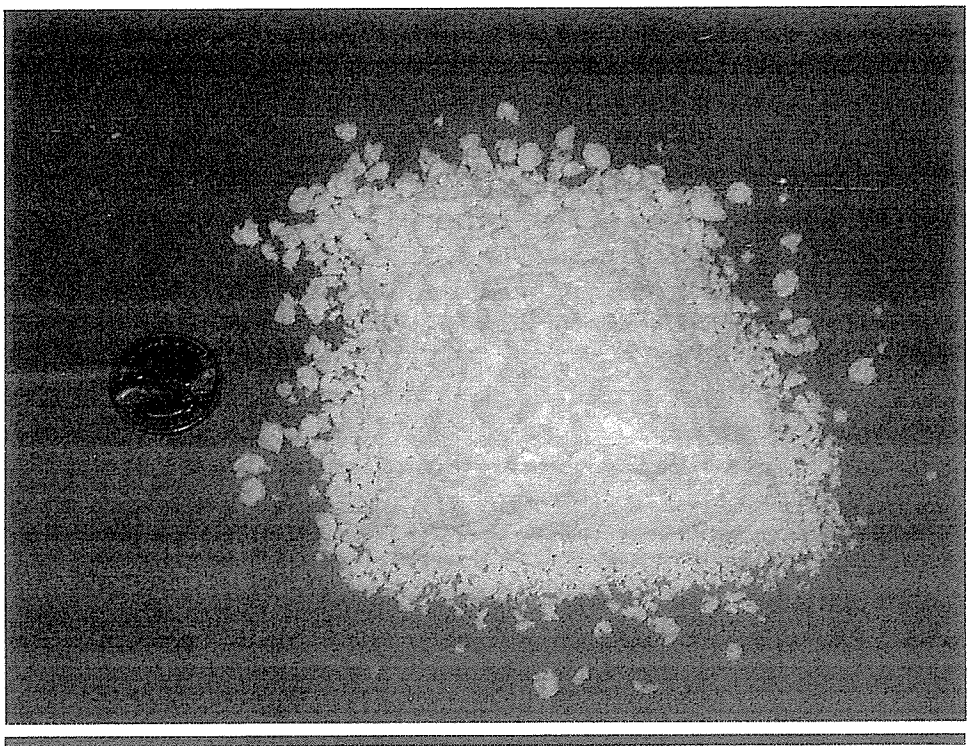


Figure 2.44 SBS Pellets

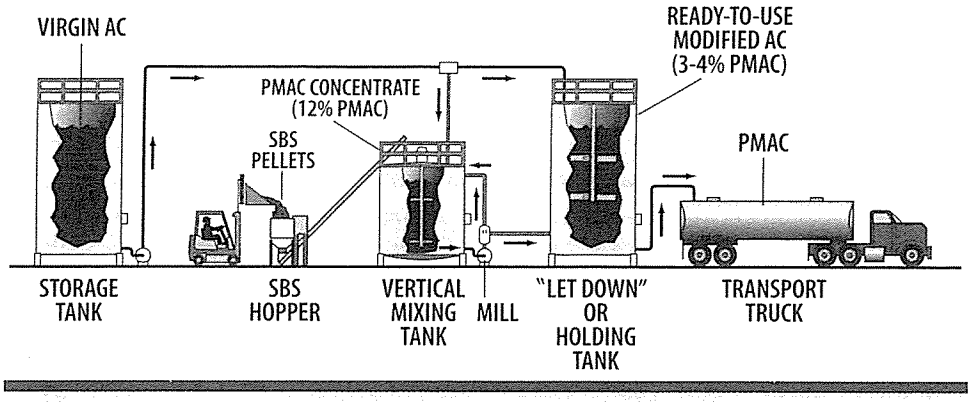


Figure 2.45 SBS Blending System for Asphalt Terminals

Surveys conducted during the NCHRP 9–10 research (entitled "Superpave Protocols for Modified Asphalt Binders") in 1996 indicated that elastomers, and more specifically SBS polymers, were the most commonly used modifiers by user agencies. In their responses, most users indicated that they selected elastomeric modifiers to help mitigate permanent deformation or rutting of asphalt pavements. Mitigation of fatigue and low-temperature cracking was also seen as a potential benefit.

In the past, specifications in the United States have sought to characterize the elastic properties of a styrenic polymer-modified asphalt binder through the use of tests that stretch the asphalt binder and measure the stress-strain response (as in the Force Ductility test and the Toughness and Tenacity test) or the recovery (as in the Elastic Recovery test). The Softening Point test has often been used to evaluate high-temperature resistance to deformation. As the percentage of styrenic polymer increases in the asphalt binder, it is expected that the high-temperature stiffness will increase as will the stress-strain and recovery response.

Plastic Modification (Plastomers)

Plastomers are a class of polymers used in asphalt binders that, when stretched, will yield and remain in their stretched position when the load is released. These polyolefin polymers add to the high-temperature stiffness of the asphalt binder but will not have the elastic characteristics of styrenic polymers. Polyolefin polymers that are commonly used to modify asphalt binders include polyethylene (often the low density polyethylene identified as LDPE) and ethylene vinyl acetate (EVA). In this class of polymers, EVA has been most commonly used in the United States.

As with elastomeric modifiers, most users indicated in the survey that they selected plastomer modifiers to help mitigate permanent deformation or rutting of asphalt pavements.

Crumb Rubber Modification

Although it is an elastic material, crumb rubber—produced from ground tires—is considered differently. Crumb rubber is incorporated in an asphalt

mixture either as an aggregate modifier (the "dry process") or as a modifier to the asphalt binder. In keeping with the objective of this chapter, only modification of the asphalt binder will be considered.

Two principal methods exist for incorporating crumb rubber as a modifier to the asphalt binder. The wet process (also referred to as the "McDonald process" named for the first patented technology that applied the wet process) is conducted by pre-blending and reacting crumb rubber of a relatively small size (usually 600 microns or smaller) in the asphalt binder at an elevated temperature for a time.

The terminal blend process is a more recent development in the crumb rubber modification of asphalt binders. In this process, a smaller volume of smaller crumb-rubber particles is continuously blended with the asphalt binder, thus eliminating the need for reaction tanks.

In the United States, crumb-rubber modification is used extensively by some users and not at all by other users. Generally, crumb-rubber modification is considered to address the same distresses (permanent deformation) as elastomeric modification.

Mixed Modification

Mixed-polymer modification, such as the combination of styrenic and polyolefin polymers, is not commonly used. Potential chemical incompatibility and the cost of using two polymers in the asphalt modification process may make the combination undesirable. However, it is possible that the addition of an elastomeric polymer to a plastomeric modifier could enhance low-temperature characteristics while maintaining high-temperature stiffness.

Chemical Modification

In this section, chemical modification refers to the addition of polyphosphoric acid (PPA) to an asphalt binder, either alone or in combination with other types of polymer modification. In general, polyphosphoric acid increases the high-temperature stiffness of an asphalt binder. However, the effect depends on the concentration and the source (properties) of the crude used to produce the asphalt binder.

Extenders

Asphalt binder extenders include materials such as sulfur and wood lignin. Of the two, sulfur has been the most extensively used and evaluated in asphalt mixtures in the United States—primarily from 1973 to 1985. At that time, asphalt binder and sulfur prices created a favorable situation enabling users to consider substitution of asphalt binder with sulfur. Although laboratory and field studies indicated a general increase in the high-temperature stiffness of asphalt mixtures incorporating sulfur modification, economic considerations eventually resulted in its reduced use.

Oxidants and Antioxidants

Oxidants principally include manganese compounds and are used to artificially "oxidize" the asphalt binder, thereby increasing its stiffness. These types of materials were used experimentally in the United States in the 1980s, but have not been commonly used since.

Antioxidants serve the opposite purpose, in that they are intended to minimize the oxidative hardening of asphalt binders. Most typically, these types of materials include carbamates (lead, zinc), carbon black, and hydrated lime. Of this class of materials, hydrated lime is the most common modifier. However, most users select hydrated lime for its benefits in reducing the moisture susceptibility of asphalt mixtures irrespective of any antioxidant benefits it may have.

Hydrocarbons

Hydrocarbon modifiers generally fall into two generic categories: hardening modifiers and softening (or rejuvenating) modifiers.

In the former category are natural asphalts and refined asphalts. Natural asphalts that have been used as modifiers include Gilsonite (a zero-penetration resin) and Trinidad Lake Asphalt (TLA). Gilsonite is added as a dry additive to either the asphalt binder or mixture. TLA is slightly softer than Gilsonite and is similar to some industrial, air-blown asphalts, except that it contains a significant amount of inert organic matter. Refined asphalts may include solvent de-asphalted bottoms such as ROSE unit asphaltenes. In either case, hydrocarbon hardening modifiers are used to increase the high-temperature stiffness (resistance to permanent deformation) of the asphalt binder.

Softening or recycling modifiers include napthenic or aromatic oils. These modifiers may be added to the asphalt binder or to the asphalt mixture separately from the asphalt binder. Most often, they are used in asphalt mixtures containing reclaimed (or recycled) asphalt pavement (RAP) as they serve to lower the viscosity of aged asphalt binders.

Antistripping Additives

Antistripping additives are used to reduce or eliminate the moisture susceptibility of an asphalt mixture by promoting adhesion between the asphalt binder and aggregate surface. Liquid antistripping additives principally include amines such as polyamines and fatty amidoamines. Most often, the practice is to blend the liquid antistripping additive with the asphalt binder before mixing with aggregate. The antistripping concentration is typically 0.5 percent of the total weight of the asphalt binder. A slight decrease in viscosity may occur with the addition of the additive (which is fluid at room temperature).

As discussed earlier, hydrated lime is the most common dry antistripping additive. Typically, hydrated lime is added at a concentration of 1 percent by weight of the dry aggregate. It may be added in the following ways: dry to the mixer, dry to wet aggregate, or as a slurry to aggregate. Users have found benefits and disadvantages with each procedure. In addition to its effectiveness as an antistripping additive, some technologists consider that it may also provide antioxidant benefits to the asphalt mixture.